

Exploratory Data Analysis (EDA) Report – CreditPathAI Project

1. Introduction

Exploratory Data Analysis (EDA) is the process of analyzing and understanding the dataset before applying machine learning models. It helps identify patterns, missing values, data distribution, and relationships between variables. In the CreditPathAI project, EDA is performed to understand customer financial behavior and identify factors that may affect credit risk and loan repayment.

2. Dataset Description

The dataset contains financial, demographic, and behavioral information about loan applicants. Each row represents a borrower, and the columns represent attributes that may affect loan approval and repayment capability.

Dataset Features

Column	Description
LoanID	Unique identifier for each loan
Age	Age of the borrower
Income	Annual income of the borrower
LoanAmount	Amount of loan taken
CreditScore	Credit score representing creditworthiness
MonthsEmployed	Number of months the borrower has been employed
NumCreditLines	Number of active credit lines
InterestRate	Interest rate applied to the loan
LoanTerm	Duration of the loan
DTIRatio	Debt-to-Income ratio
Education	Educational qualification of borrower
EmploymentType	Type of employment
MaritalStatus	Marital status of borrower
HasMortgage	Indicates if borrower has a mortgage
HasDependents	Indicates if borrower has dependents
LoanPurpose	Purpose of the loan

HasCoSigner	Indicates if the loan has a co-signer
Default	Target variable indicating whether the borrower defaulted

3. Data Cleaning

Before performing exploratory analysis, the dataset was preprocessed to ensure accuracy and consistency.

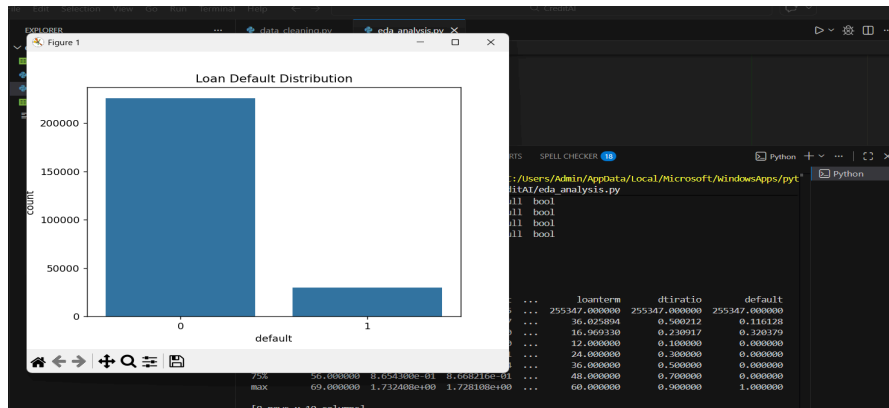
First, the dataset was checked for missing values, and it was found that no missing values were present in any column. This ensured that the dataset was complete and did not require imputation. Next, the dataset was examined for duplicate records, and no duplicate rows were detected, ensuring data integrity. Categorical attributes such as Education, EmploymentType, MaritalStatus, and LoanPurpose were reviewed and standardized to maintain consistency in the dataset. Boolean attributes including HasMortgage, HasDependents, and HasCoSigner were converted into True/False format to simplify processing and analysis. These preprocessing steps ensured that the dataset was clean, consistent, and ready for exploratory analysis.

4. Exploratory Analysis

Exploratory analysis was conducted to understand the distribution and relationships among different variables in the dataset.

Key aspects analyzed include:

- Distribution of borrower age and income
- Analysis of loan amounts and loan terms
- Examination of credit scores and debt-to-income ratio
- Relationship between financial factors and loan default
- Distribution of categorical variables such as education and employment type



5. Key Insights

Some important observations from the exploratory analysis include:

- Borrowers with lower credit scores show a higher probability of loan default.
- Higher Debt-to-Income ratios may indicate increased financial risk.
- Individuals with stable employment and higher income tend to have lower default rates.
- Loan purpose and number of credit lines may influence borrower repayment behavior.

6. Conclusion

The exploratory data analysis provided a deeper understanding of borrower financial characteristics and their potential impact on loan repayment behavior. The dataset was successfully cleaned and analyzed, and it was confirmed that no missing values or duplicate records were present. The analysis revealed meaningful patterns related to credit risk and borrower profiles.